Ayaan Ahmad

Do Cars Need Advanced Technology?

Exploring cognitive load, physical controls and the role of technology in modern automotive HMI.

Earlier, cars used to be where people could operate things without looking.
Then things changed.
Screens got bigger. Features multiplied. Buttons disappeared.
Functions that once took a second became menus, gestures and taps.

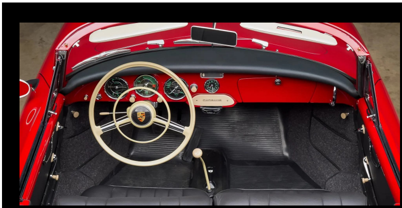
And that made me wonder:
Are cars actually becoming easier to use?
Or are we simply giving them more things to do?

That question became the starting point for this project.

THEN AND NOW

Background: How We Got Here...

1900s – 1970s

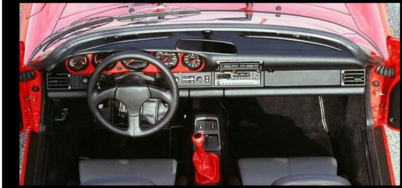


Mechanical Era

The interface was almost entirely physical and directly connected to the car's functions. Analogue gauges, levers and switches offered immediate feedback with minimal cognitive effort.

Porsche 356

1980s–2000s



Button Era

As cars gained more functions, dedicated buttons and knobs became the dominant interaction method. Each function had a clear physical control, encouraging consistency and muscle memory.

Porsche 911 (964)

1998–2004

Early Digital Era

Electronics began entering the cockpit without completely replacing physical interaction. Digital displays and early infotainment added new capabilities while traditional buttons, knobs and analogue instruments remained prominent.

Porsche 911 (996)

2009–2024

Screen Era

Touchscreens increasingly became the central interface, allowing manufacturers to consolidate many functions into a single digital layer. This brought flexibility and visual richness, but also moved more interactions behind screens and menus.

Porsche Panamera (970)

2024–present

Rebalancing Era

The industry is now moving toward a hybrid approach, recognising that not every function benefits from being digital. Screens handle dynamic and information-rich tasks, while frequently used or critical functions remain immediately accessible.

Porsche Panamera (3rd generation)

The Problem

As vehicles become increasingly software-defined, more functions are being moved from dedicated physical controls into touchscreens and layered digital interfaces.
While this enables greater functionality, flexibility and visual richness, it can also increase the driver's visual, manual and cognitive workload.

Main Research Question:
Is increasing technological complexity in cars actually improving the driving experience?

01 — Cognitive load
How does the way a driver interacts with a vehicle affect cognitive workload and visual attention?
02 — Modality
Which functions benefit from physical controls, touchscreens, voice or other interaction methods?
03 — Context
How should the interface behave when the driver's attention is already occupied by the road?

The downfall of society in two photos
Then
Now

Project Objective

The objective of this study is to understand how automotive HMI design affects driver cognitive load and to identify when technology genuinely improves the driving experience versus when it introduces unnecessary interaction.

OUR CAR JOURNEY
FAMILY TRIPS
LONG DRIVES
MEMORIES UNFOLDED
UNEXPECTED ADVENTURES

A Cinematic Documentary on our Car Journey since 2011.

My relationship with cars started long before I began studying their interfaces.
Since 2011, I've experienced the shift from simple, physical controls to increasingly screen-driven interiors—
not just as a designer, but as a passenger and a driver.

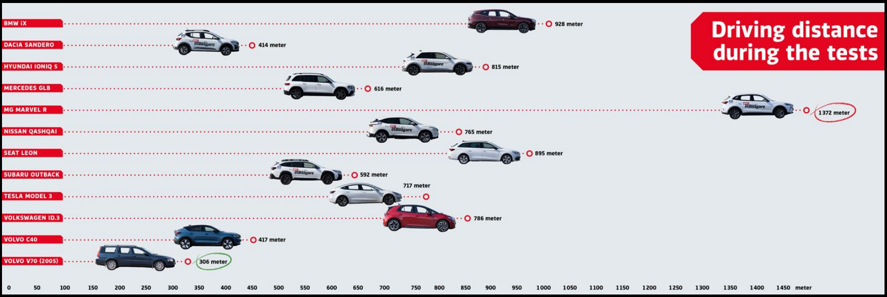
This personal journey made me question something:
as cars become more technologically capable, are they also becoming more mentally demanding to use?

What the industry is already telling us – SECONDARY RESEARCH.

Physical buttons outperform touchscreens in new cars, test finds

By Vi Bilägare | 2022

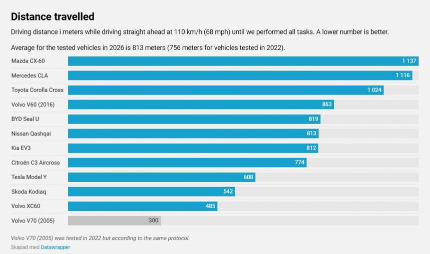
In 2022, they tested twelve cars while drivers performed basic tasks like adjusting the climate, changing the radio, and controlling the display — all while driving at 110 kilometres per hour.
The results were striking. The 2005 Volvo V70 completed the tasks in just 10 seconds, while the MG Marvel R took almost 45 seconds — meaning the driver travelled more than four times the distance.



Best Performer: 2005 Volvo V70 - 10.0 sec | 306 m

Worst Performer: MG Marvel R - 44.9 sec | 1,372 m

But their 2026 follow-up revealed something even more interesting.
The problem isn't simply touchscreens versus physical buttons.



Distance travelled

Driving distance i meters while driving straight ahead at 110 km/h (68 mph) until we performed all tasks. A lower number is better.

Average for the tested vehicles in 2026 is 813 meters (756 meters for vehicles tested in 2022).

Volvo V70 (2005) was tested in 2022 but according to the same protocol.

Skapad med Datawrapper

Average interaction distance actually increased from 756 to 813 metres.
Yet the Volvo XC60, with a screen-heavy interface, performed the best.

When HMI Becomes a Safety | Issue Euro NCAP — 2026 Safety Protocol

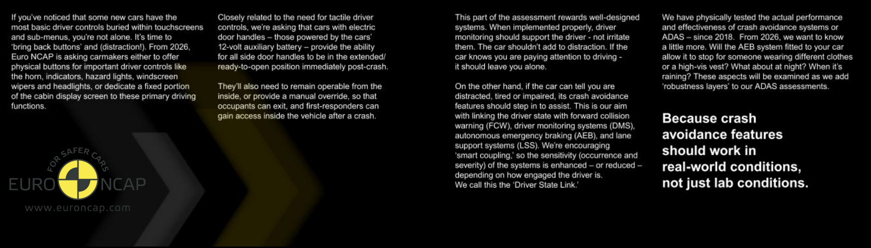
By Euro NCAP | 2026

Physical Buttons

Electric Door Handles

Driver Engagement

ADAS Robustness



If you've noticed that some new cars have the most basic driver controls buried within touchscreens and sub-menus, you're not alone. It's time to 'bring back buttons' and (distraction!). From 2026, Euro NCAP is asking carmakers either to offer physical buttons for important driver controls like the horn, indicators, hazard lights, windscreen wipers and headlights, or dedicate a fixed portion of the cabin display screen to these primary driving functions.

Closely related to the need for tactile driver controls, we're asking that cars with electric door handles – those powered by the cars' 12-volt auxiliary battery – provide the ability for all side door handles to be in the extended/ready-to-open position immediately post-crash.

They'll also need to remain operable from the inside, or provide a manual override, so that occupants can exit, and first-responders can gain access inside the vehicle after a crash.

This part of the assessment rewards well-designed systems. When implemented properly, driver monitoring should support the driver - not irritate them. The car shouldn't add to distraction. If the car knows you are paying attention to driving - it should leave you alone.

On the other hand, if the car can tell you are distracted, tired or impaired, its crash avoidance features should step in to assist. This is our aim with linking the driver state with forward collision warning (FCW), driver monitoring systems (DMS), autonomous emergency braking (AEB), and lane support systems (LSS). We're encouraging 'smart coupling,' so the sensitivity (occurrence and severity) of the systems is enhanced – or reduced – depending on how engaged the driver is. We call this the 'Driver State Link.'

We have physically tested the actual performance and effectiveness of crash avoidance systems or ADAS – since 2018. From 2026, we want to know a little more. Will the AEB system fitted to your car allow it to stop for someone wearing different clothes or a high-vis vest? What about at night? When it's raining? These aspects will be examined as we add 'robustness layers' to our ADAS assessments.

Because crash avoidance features should work in real-world conditions, not just lab conditions.

"Essential controls should be accessible without navigating complex touchscreen menus."

"Touch Controls Were The Future. Until They Weren't"
"EV Buyers Still Want Familiar Controls"

"People get frustrated when simple controls aren't quick and easy to access; they preferred physical controls over touchscreen"

ACCESSIBILITY > COMPLEXITY
The driver shouldn't have to search for something they need now.

The message is becoming clear:
The future of automotive HMI isn't about having fewer screens.
it's not the technology that matters — it's how well the interface is designed.

ADAS in cars in India: Is it useful or not?

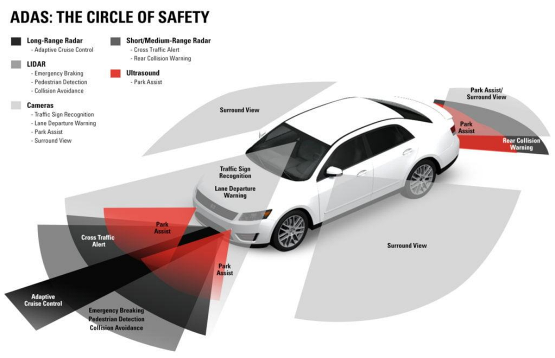
By Team-BHP | 2023

The technology works. The context is different.

India's roads create a different set of challenges:

- Unclear lane markings
- Unpredictable two-wheeler movement
- Dense, mixed traffic
- Frequent road-edge activity
- Informal driving behaviour

ADAS can be highly useful on controlled highways, but Indian city traffic can make systems like AEB and lane assistance intrusive because of unpredictable road users and conditions.



Good technology still needs to understand the road it was designed for.

What the CAR EXPERTS have to say – PRIMARY RESEARCH.

Ishteyaaq Ahmad
Age – 50
Dehradun, UK

Rajiv Chowdhury
Age – 57
Guwahati, AS

Andrew Howard
Age – 21
Dehradun, UK

Mudassir Shah
Age – 22
Khamgaon, MH

Chetan Bokde
Age – 41
Nagpur, MH

Mayank Chandak
Age – 21
Raipur, CG

Ankush Nagwekar
Age – 23
Mumbai, MH

Aditya Rudrabhatla
Age – 22
Hyderabad, TG

Ansh Nishane
Age – 22
Nagpur, MH

Amaan Ahmad
Age – 10
Dehradun, UK

Anonymous

Anonymous

For a descriptive research documentary, watch this video:

AUTOMOTIVE HMI
BUTTONS → SCREENS
TOO TECHNICAL?
Know what Car Experts Say!
TECHNOLOGY IS CHANGING CARS

Here’s what Car Experts said:

"Technical advancement in cars is necessary until it's not overcomplicating the function or the use of that particular thing."

"Technology is important but I feel it should have a purpose rather than just being added in it."

"We should fully be concerned on the driving. If we are distracted then we'll ourselves be out of control."

What I Learned from the Car Experts

Physical controls can reduce the need to look away from the road.

Complexity can make advanced technology frustrating to use.

Digital features are valuable when they solve everyday problems.

Drivers want confidence, feedback and a sense of control.

Technology must respond to real-world driving conditions.

A good driving experience balances convenience with human involvement.

Overall Insight

The goal is not to remove technology from cars, but to integrate it in ways that make driving safer, simpler and intuitive.

Problem Statement

As car interiors increasingly integrate digital technology, drivers can face unnecessary visual attention demands and more complicated interactions when accessing essential functions. There is an opportunity to balance digital convenience with intuitive, accessible controls in order to give the user a good driving experience.

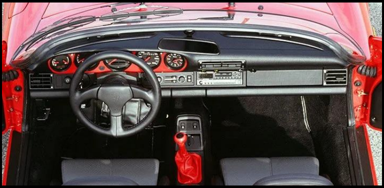
How Might We Statement

How might we design a car interior that combines the benefits of modern technology with simple, intuitive interactions that minimize driver distraction?

Solution Statement

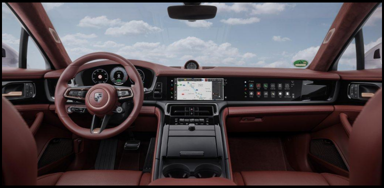
I propose to design a human-centred automotive HMI that balances modern digital technology with intuitive physical controls, keeping essential functions easy to access, reducing unnecessary visual distraction, and preserving the driver's sense of control. The goal is to create a clean, purposeful car interior where technology enhances the driving experience rather than complicating it.

What's the difference?



Button Era - Porsche 911 (964)

Dedicated buttons and knobs gave drivers direct control over

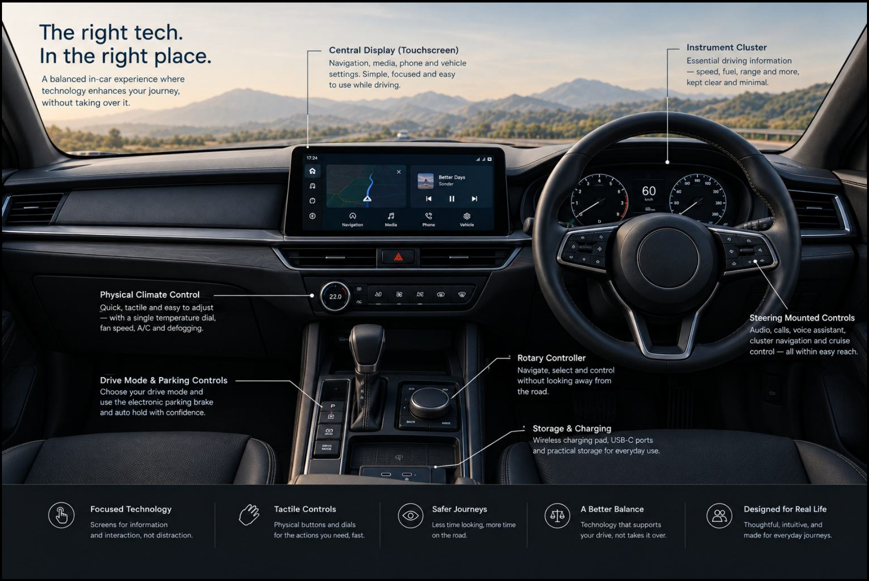


Rebalancing Era - Porsche Panamera (3rd generation)

Physical controls are returning for essential functions, while individual functions, making frequent actions easier to access by feel. However, as features increased, crowded dashboards and numerous similar-looking controls could make certain functions confusing to locate and operate.

digital screens handle navigation, infotainment and other advanced features. However, with digital interfaces still dominating much of the dashboard, large screens and layered menus can be distracting and difficult to operate while driving.

The solution lies somewhere in between -
an intuitive car interior that brings physical and digital controls together in the right balance. A design where essential functions are effortless to access, digital technology serves a clear purpose, and the driver can focus on the road rather than figuring out the interface.



The right tech.
In the right place.

A balanced in-car experience where technology enhances your journey, without taking over it.

Central Display (Touchscreen)
Navigation, media, phone and vehicle settings. Simple, focused and easy to use while driving.

Instrument Cluster
Essential driving information — speed, fuel, range and more, kept clear and minimal.

Physical Climate Control
Quick, tactile and easy to adjust — with a single temperature dial, fan speed, A/C and defogging.

Steering Mounted Controls
Audio, calls, voice assistant, cluster navigation and cruise control — all within easy reach.

Rotary Controller
Navigate, select and control without looking away from the road.

Drive Mode & Parking Controls
Choose your drive mode and use the electronic parking brake and auto hold with confidence.

Storage & Charging
Wireless charging pad, USB-C ports and practical storage for everyday use.

Focused Technology
Screens for information and interaction, not distraction.

Tactile Controls
Physical buttons and dials for the actions you need, fast.

Safer Journeys
Less time looking, more time on the road.

A Better Balance
Technology that supports your drive, not takes it over.

Designed for Real Life
Thoughtful, intuitive, and made for everyday journeys.

This AI generated image represents a possible direction of what an ideal car interior looks like...
but an intuitive car interior requires more than a visual balance between physical and digital controls.

My research goes deeper into how users interact with these systems, how different controls affect cognitive load, and which functions deserve immediate physical access versus a digital interface.

The goal is to go beyond how the dashboard looks and understand how it can work better for the person behind the wheel.

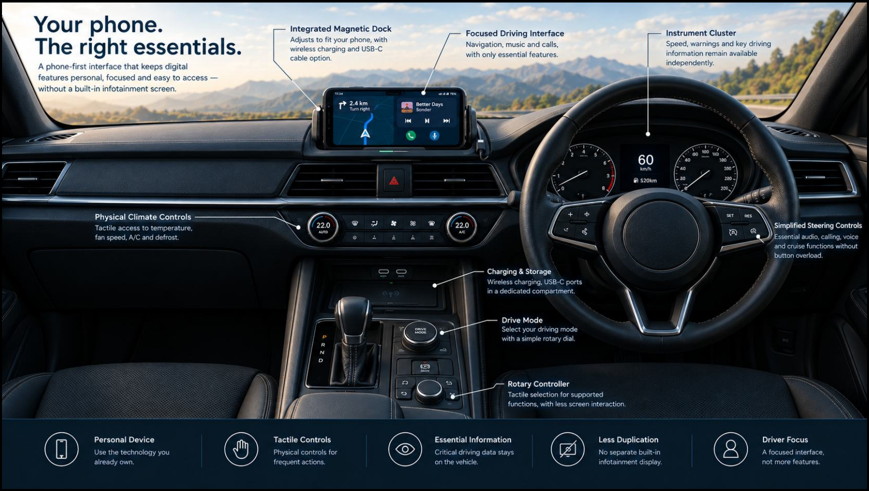
ROUND 01 — INITIAL HMI USER FEEDBACK

I presented my initial automotive HMI concept to car experts to understand their perspectives on accessibility, control placement and the overall driving experience.

Key feedback themes:

- Accessibility: Drive-mode buttons and essential controls needed better placement.
- Cognitive Load: Excessive steering-wheel buttons and overlapping screen menus could make interactions more difficult.
- Practicality: Charging ports, storage, handbrake placement and screen repairability needed reconsideration.
- A new direction: Rather than creating another conventional infotainment system, I was encouraged to explore a more original approach.

But that is just something generic, right?
Why don’t we think of something out of the box?



Your phone.
The right essentials.

A phone-first interface that keeps digital features personal, focused and easy to access — without a built-in infotainment screen.

Integrated Magnetic Dock
Adjusts to fit your phone, with wireless charging and USB-C cable option.

Focused Driving Interface
Navigation, music and calls, with only essential features.

Instrument Cluster
Speed, warnings and key driving information remain available independently.

Physical Climate Controls
Tactile access to temperature, fan speed, A/C and defrost.

Simplified Steering Controls
Essential audio, calling, voice and cruise functions without button overload.

Charging & Storage
Wireless charging, USB-C ports in a dedicated compartment.

Drive Mode
Select your driving mode with a simple rotary dial.

Rotary Controller
Tactile selection for supported functions, with less screen interaction.

Personal Device
Use the technology you already own.

Tactile Controls
Physical controls for frequent actions.

Essential Information
Critical driving data stays on the vehicle.

Less Duplication
No separate built-in infotainment display.

Driver Focus
A focused interface, not more features.

*applied changes after round 01 testing / recommendations by car enthusiasts

Let’s check out something NEW...

PORSCHE
STUTTGART

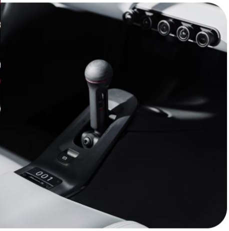
Mission S

NOT ANOTHER CAR

A SPORTS CAR.

482
251
NORMAL
4.8

107
SPORT
4.8



FOCUS ON WHAT MATTERS.

Every idea behind the Mission S stems from a single question: what does the sports car of tomorrow really need? The answer is a consequent focus on the driver experience. From the five round instruments with an analogue rev counter to the centre dials with push-pull function.

Inspired by Porsche. Designed around the Driver.

The Porsche Mission S explores a different approach to automotive interfaces by using a smartphone as the primary digital display instead of a permanently integrated infotainment screen.

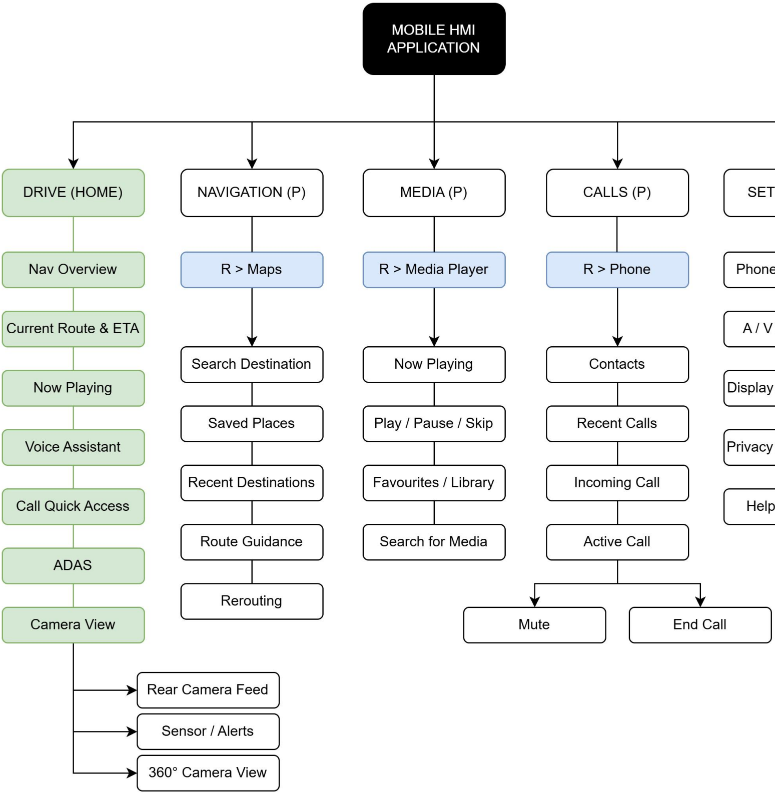
Inspired by this direction, I explored how a phone-first HMI could simplify the dashboard while keeping essential driving functions intuitive and accessible.

My focus is on more than removing a screen. It is about giving every function the right place, reducing unnecessary cognitive load, and ensuring technology serves the driver rather than distracting them.

It has no permanent **centre touchscreen.**

Instead of a **built-in centre screen**, Porsche designed a retractable **MagSafe**-compatible phone mount for **navigation and information.**

Now, let's actually "Focus on What Matters."



- MOBILE HMI APPLICATION
  - DRIVE (HOME)
    - Nav Overview
    - Current Route & ETA
    - Now Playing
    - Voice Assistant
    - Call Quick Access
    - ADAS
    - Camera View
      - Rear Camera Feed
      - Sensor / Alerts
      - 360° Camera View
  - NAVIGATION (P)
    - R > Maps
      - Search Destination
      - Saved Places
      - Recent Destinations
      - Route Guidance
      - Rerouting
  - MEDIA (P)
    - R > Media Player
      - Now Playing
      - Play / Pause / Skip
      - Favourites / Library
      - Search for Media
  - CALLS (P)
    - R > Phone
      - Contacts
      - Recent Calls
      - Incoming Call
      - Active Call
        - Mute
        - End Call
  - SETTINGS (P)
    - Phone Connection
    - A / V Preference
    - Display / Accessibility
    - Privacy / Permissions
    - Help & Support

Bringing the Concept to Life

Moving beyond wireframes, I developed an interactive phone-first HMI prototype that brings together essential driving features such as navigation, music, calls, voice assistance and vehicle information.

The prototype demonstrates how a simplified, context-aware interface can reduce unnecessary interactions while keeping the driving experience focused, intuitive and connected.

phone_first_hmi_prototype_package

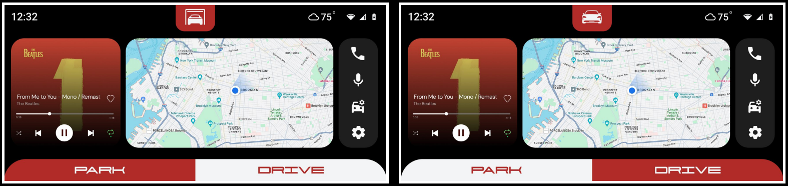
Taking Porsche as the Benchmark

COLORS OF
PORSCHE®

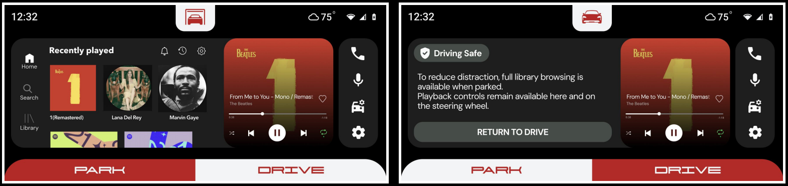
#000000 **#191C1E** **#F3F4F5** **#B12B28** **#EBD698** **#464C47**

Designing the Mobile HMI as if it were in a Porsche Car

Ready for the Actual UI Design?

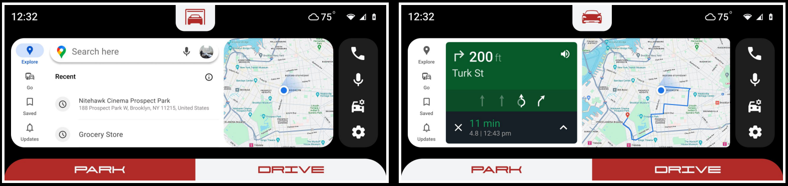


A minimal home screen that brings together Park and Drive modes, navigation, media and essential vehicle features for quick, intuitive access.

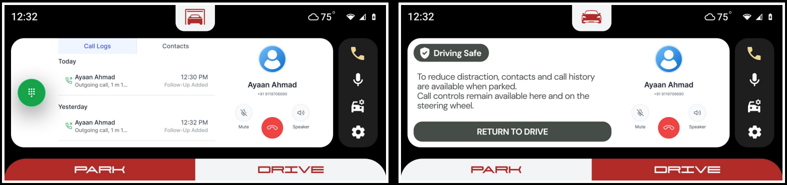


To reduce distraction, full library browsing is available when parked.
Playback controls remain available here and on the steering wheel.

Selecting Media opens the user's preferred music app, such as Spotify or Apple Music, with restricted browsing and simplified playback controls while driving.

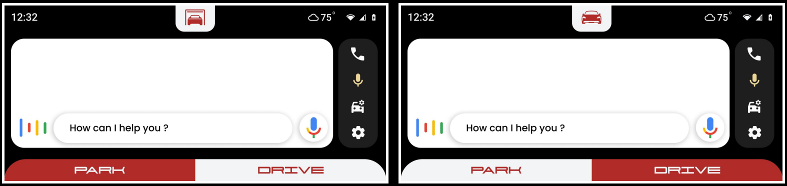


Selecting Map opens the user's preferred navigation app, such as Google Maps, with simplified, easy-to-follow guidance and limited interaction while driving.

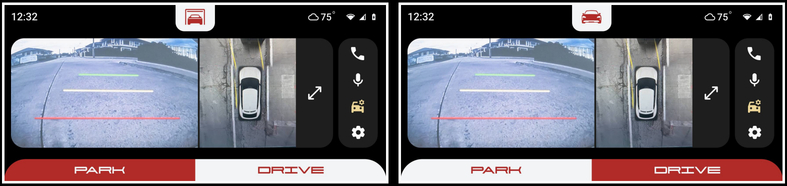


To reduce distraction, contacts and call history are available when parked.
Call controls remain available here and on the steering wheel.

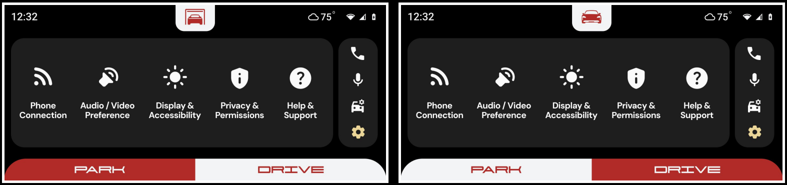
Selecting the Phone icon opens the calling interface, providing quick access to essential communication features such as contacts, recent calls, and call controls, with limited interaction while driving.



Activating the microphone launches an AI-powered voice assistant, enabling drivers to manage navigation, media, calls and supported vehicle functions through natural voice commands.



Selecting the Vehicle Features icon opens the rear and 360° camera views,
helping drivers stay aware of their immediate surroundings while parking and reversing



The Settings section provides access to essential vehicle and device preferences, including connectivity, audio and video, display, permissions and help.

Mobile HMI Application UI Prototype

ROUND 2 — VALIDATING THE PHONE-FIRST HMI

After developing the phone-first automotive HMI and its interactive mobile interface, I conducted a second round of testing with eight participants from my earlier research.

The feedback was encouraging, with participants appreciating the simplified interface, direct use of personal smartphones and the distinction between Park and Drive modes. However, testing also revealed areas that require further consideration before the concept can become a practical solution.

Key User Insights –

01 — Device Compatibility & Accessibility

Phone sizes, mounting adjustments, visibility at night and text readability emerged as important considerations. A universal, adjustable mounting mechanism and adaptable interface sizing would be necessary.

02 — Driving Safety & Interaction

Participants appreciated the Park and Drive separation and the restricted driving interface. Voice assistance should remain accessible throughout the journey, while complex browsing and other distracting interactions should remain restricted.

03 — Vehicle Integration

Reliable integration with the vehicle's microphone, speakers, rear camera, 360° camera and other systems needs further exploration. Emergency calling and favourite contacts were also suggested as useful additions.

04 — Interface Simplicity

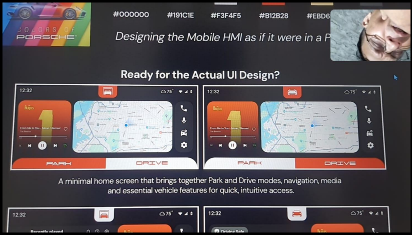
The minimal interface and direct use of personal applications were positively received. Participants appreciated avoiding unnecessary Android Auto-style intermediary interactions and having only essential functions readily accessible.

05 — Practicality & Cost

The phone-first approach was recognised as a potential way to reduce dedicated infotainment hardware, screen duplication and manufacturing complexity. A magnetic mounting and charging mechanism was also proposed.

06 — Adoption & Market Suitability

Feedback revealed that not every driver would prefer a screenless dashboard. Some participants valued the permanence and larger viewing area of a built-in infotainment display, suggesting that the concept could work better as an alternative configuration rather than a universal replacement.



Designing the Mobile HMI as if it were in a P

Ready for the Actual UI Design?

A minimal home screen that brings together Park and Drive modes, navigation, media and essential vehicle features for quick, intuitive access.

WHAT THE TESTING REVEALED

The second round of testing indicated that the phone-first HMI is a promising alternative to conventional infotainment systems. Participants appreciated its minimal interface, direct use of personal applications, simplified driving mode and separation of essential vehicle controls from digital interactions.

At the same time, the feedback highlighted practical challenges around device compatibility, screen visibility, vehicle integration and user preferences.
The concept received positive usability feedback, but its success depends on making the experience adaptable to different devices, drivers and vehicle environments.

DESIGN REFINEMENTS IDENTIFIED

Based on the second round of testing, the following improvements were identified:

1. Explore a universal, adjustable and secure smartphone mounting mechanism.
2. Introduce adaptable text sizing and automatic day/night display behaviour.
3. Keep voice-assistant access consistently available throughout the drive.
4. Explore emergency calling and favourite-contact shortcuts.
5. Ensure reliable integration with vehicle cameras, microphone and audio systems.
6. Investigate optional instrument-cluster integration for selected media information.
7. Consider the phone-first interface as an alternative to conventional built-in infotainment rather than assuming it suits every driver.

These refinements provide a direction for further prototyping and usability validation.

12:32
75°
From Me to You
PARK
DRIVE
60
km/h
520km
22.0
AUTO
22.0
A/C
SET
RES
DRIVE
MODE
P

THE FINAL THOUGHT

Is more technology really necessary?

This project began with a question about whether increasing technology in cars was genuinely improving the driving experience.

Through research, user interviews, multiple design iterations and two rounds of testing, I realised that the problem was never technology itself, but how it is introduced and experienced by the driver.

The phone-first HMI explores an alternative approach by removing the need for a permanently integrated infotainment display while retaining the digital features drivers already use. It brings together personal technology, essential vehicle functions and a simplified interaction model.

The feedback also made one thing clear: a screenless approach may not be ideal for everyone. Device compatibility, visibility, integration and personal preferences all influence the experience.

My final takeaway is that the future of automotive HMI should not be defined by how much technology a car contains, but by how intelligently that technology serves the person behind the wheel.

The goal is not to add more features or remove every screen. It is to make every interaction purposeful, intuitive and worthy of the driver's attention.

AYAAN AHMAD